Igudar AI

100% Local AI Intelligence for Fractional Real Estate Investment

Semantic Search · RAG Chatbot · Risk Assessment · ROI Prediction

Built with privacy-first architecture — zero external API dependencies

CORE TECHNOLOGIES

The Engines Powering Igudar AI

Igudar AI relies on a carefully selected stack of localized technologies that prioritize privacy, speed, and accuracy over costly cloud APIs.

1. all-MiniLM-L6-v2 (Embedding Model)

This is the foundation of our semantic understanding. Instead of matching basic keywords, MiniLM converts any text—whether it's a user's quiz answers or a property description—into a mathematical representation of its meaning, known as a "vector" (a list of 384 numbers). Because it runs 100% locally and is incredibly lightweight (only 80MB), it executes in millisecor without exposing user data.

2. pgvector (Vector Database)

pgvector is an advanced extension for our PostgreSQL database that allows us to store and instantly search the vectors generated by MiniLM. It calculates the "cosine similarity" between what the user wants and what properties offer. This completely solves the "cold start" problem we don't need thousands of previous user transactions to make highly accurate recommendations on day one.

3. Qwen2.5-0.5B-Instruct (Chat LLM)

Our localized conversational AI. Qwen is a highly optimized (500MB) language model that generates natural, human-like responses while running purely on CPU power. We use it strictly Retrieval-Augmented Generation (RAG). Before answering a user, Qwen reads the specific property data retrieved by pgvector, ensuring its answers are accurate, grounded entirely in ou actual database, and completely free from AI hallucinations.

FINANCIAL INTELLIGENCE

Evaluation & Forecasting

4. Risk Assessment Engine

A deterministic, transparent mathematical model that evaluates the risk of a property tailored specifically to the individual user. It calculates a risk score (0-100) based on four factors: the property's base risk, the lock-in period penalty, the yield premium, and the alignment with the user's stated risk tolerance. It also applies an experience modifier, meaning a high-yield prope is correctly flagged as riskier for a beginner than an expert.

5. ROI Prediction Engine

Our upcoming financial forecasting module. While the recommendation engine focuses on answering "which property," the ROI prediction answers "what are the financial outcomes." It takes a user's specific investment amount and projects their total returns over their desired holding period, factoring in the property's annual rental yield and expected capital appreciatior provide clear financial foresight.

How It All Connects

These components seamlessly work together across two independent, privacy-first pipelines:

Pipeline A: Smart Recommendation

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User Quiz → MiniLM → pgvector → Risk/ROI Engines
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The user's quiz preferences are embedded by MiniLM. Next, pgvector finds the best semantic property matches. Finally, the Risk and ROI engines attach personalized safety scores and financial forecasts to those properties before presenting them to the user.

Pipeline B: RAG Chatbot

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User Message → MiniLM → pgvector → Qwen LLM
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> The user asks a question, which is embedded by MiniLM. Then, pgvector retrieves the most relevant real properties from our database. This factual data is injected into the Qwen LLM, which generates a natural, conversational, and highly accurate reply.